Palm OS Programming Recipes

Exchange Manager: Sending Text to Local PIM Applications in Palm OS™ 3.0 or Later

Starting with Palm OS™ version 4.0, the Exchange Manager works in conjunction with an exchange library. Each exchange library is transport-dependent and performs the actual communication with the remote device.

In Palm OS version 3.x, you need to set the `localMode` flag to `1` to interact with the local device using the IR Library (the only communications library available in Palm OS 3.x). If you want to use the Exchange Manager to exchange data with other applications on the local device in Palm OS version 4.0, you can use the Local Exchange Library. Please refer to other recipes for details.

The following recipe shows you how to send a message to the local Memopad application using 3.0 or later. It will send a string buffer, with the value "test," as a .txt data type to Memopad. You should be focused on how to set up the data structure `ExgSocket`, which is the core of the Exchange Manager.

Follow these steps:

In Constructor:

- Add a menu item for sending or for any other UI components that are applicable to your application.

Open the source file (`*.c`) and find the event loop handling function that will handle the "Send" menu item. Modify it as follows:

- Make sure that `ExgMgr.h` is included as a header file.
- Create and initialize an `ExgSocketType` data structure with information about which library to use and the data to be sent. Make sure to initialize the description, type, target, and/or name.
- Set `localMode` to 1 for `ExgSocketType` (this is the setting for sending to a local device).
- Set `name` to end with .txt. This is the file type that Memopad is registered to receive.
- Call the `ExgPut` function to establish the connection with the exchange library.
- Call the `ExgSend` function to send the actual data in a loop (if the sending buffer is big).
- Call the `ExgDisconnect` function to end the connection.

These code segments are examples of the above procedure:

```
static void SendToLocalMemopad()
{
    ExgSocketType exgSocket;
    Err err = 0;
    // message that will be sent
    // to memopad
    Char        *textBuf = "test";
    UInt32 size = StrLen(textBuf) + 1;
```

```
        // important to init structure to zeros...
        MemSet(&exgSocket,sizeof(exgSocket),0);
        exgSocket.description = "Testing message from
            ExgMgrLocalPIM";
        // will comm. with memopad app
        exgSocket.name = "Sample.txt";
        // set to comm. with local machine (PIM) only
        exgSocket.localMode = 1;

        err = ExgPut(&exgSocket);
        if (err == 0) {
      ExgSend(&exgSocket,textBuf,size,&err);
      ExgDisconnect(&exgSocket,err);
     }
    }
```

Sample program downloads:

- Windows
- Mac

See the recipe, "Sending Text to Local PIM Applications in Palm OS 4.0", for sending data locally with the Exchange Manager in later versions of the Palm OS.